

James Zhu

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Education

PhD, Mechanical Engineering

Carnegie Mellon University

September 2024

Thesis: Navigating a Complex World: Improving Robot Outcomes Through Social, Regulatory, and Control Theoretic Approaches

MS, Mechanical Engineering

Carnegie Mellon University

December 2022

BE, Mechanical Engineering and Mathematics

Vanderbilt University

May 2020

Cum Laude

Teaching Experience

University of Miami

Miami, Florida

Co-Instructor: Robotics and Intelligent Systems & Robotics Studio Lab

July 2026 – Current

- Modernized introductory robotics curriculum to incorporate modern control, robot learning, and deep learning
- Developed a hands-on curriculum for an intensive robotics studio laboratory course
- Designed and deployed custom robotic-arm platforms for hands-on student projects

Postdoctoral Researcher: Robotics & Mechatronics Lab

July 2026 – Current

- Mentored and coordinated 3 PhD, 2 Master's, and 4 undergraduate students, launching multidisciplinary projects in legged robotics, assistive robotics, and environmental monitoring
- Organized monthly "Learning Lunches" providing professional and technical development in areas including robotics, AI, responsible technology, and career preparation

Carnegie Mellon University

Pittsburgh, PA

Teaching Consultant Fellow: Eberly Center for Teaching Excellence

September 2022 – December 2023

- Provided one-on-one teaching consultations to graduate students on course design and teaching practice
- Designed and led workshops on subjects such as inclusive teaching practices and effective teaching statements
- Led pedagogy roundtables on evidence-based teaching practices and curriculum design

Founder & Organizer: Gwen's Girls Robotics Outreach Program

September 2020 – December 2023

- Partnered with Gwen's Girls to develop an after-school engineering program for middle-school students
- Designed an engineering design curriculum incorporating hands-on project-based learning
- Secured local government funding to equip program with 3D printers and establish engineering program
- Program featured in Carnegie Mellon Engineering Magazine

Other Teaching Experience

Education Volunteer: Zoo Miami

September 2026 – Present

Advisory Board Member: CMU Teaching & Learning Summit

September 2023

Inclusive STEM Teaching Certificate

April 2023

Teaching Assistant: Dynamics

Spring 2022 and Spring 2023

Teaching Assistant: Intro to Robotics

Spring 2019

Teaching Assistant: Probability and Statistical Inference

Spring 2019

Work Experience

Postdoctoral Associate

University of Miami

Miami, Florida

July 2026 – Current

Supervisor: Pinhas Ben-Tzvi

Postdoctoral Researcher

French National Center for Scientific Research

Toulouse, France

January 2025 – July 2026

Supervisors: Thierry Simeon & Marco Cognetti

Graduate Research Assistant

Carnegie Mellon University

Advisor: Aaron Johnson

Robots, Ethics, & Society Research Intern

The Robotics & AI Institute

Supervisor: Kate Darling

Undergraduate Research Assistant

Vanderbilt University

Advisors: Nilanjan Sarkar & Robert Webster

High Contrast Imaging Intern

Jet Propulsion Laboratory

Supervisor: Stuart Shaklan

Engineering and Operations Intern

AT&T

Pittsburgh, Pennsylvania

May 2020 – September 2024

Boston, Massachusetts

January 2024 – June 2024

Nashville, Tennessee

September 2017 – May 2020

Pasadena, California

May 2019 – July 2019

Atlanta, Georgia

May 2018 – August 2018

Certifications & Training

Summer Institutes in Computational Social Science Program

June 2025

Tech Stewardship Practice Program Certificate

December 2022

Languages

English: Native

French: Advanced (C2) – TCF Tout Public Certified

Mandarin Chinese: Heritage

Spanish: Beginner

Selected Publications

Journal Papers

Robust Sensitivity-Aware Chance-Constrained MPC for Efficient Handling of Multiple Uncertainty Sources

James Zhu, Thierry Simeon, and Marco Cagnetti

2025 IEEE Robotics and Automation Letters

Hybrid Iterative Linear Quadratic Estimation: Optimal Estimation for Hybrid Systems

J Joe Payne, James Zhu, Nathan J Kong, and Aaron M Johnson

2025 IEEE Robotics and Automation Letters

Saltation Matrices: The Essential Tool for Linearizing Hybrid Dynamical Systems

Nathan J Kong, J Joe Payne, James Zhu, and Aaron M Johnson

2024 Proceedings of the IEEE

Conference Papers

Convergent iLQR for Safe Trajectory Planning and Control of Legged Robots

James Zhu, J Joe Payne, and Aaron M Johnson

2024 IEEE International Conference on Robotics and Automation

Grounding Robot Navigation in Self-Defense Law

James Zhu, Anoushka Shrivastava, and Aaron M Johnson

2023 IEEE International Conference on Robot and Human Interactive Communication

Hybrid Event Shaping to Stabilize Periodic Hybrid Orbits

James Zhu, Nathan J Kong, George Council, and Aaron M Johnson

2022 IEEE International Conference on Robotics and Automation

Leadership, Honors, and Additional Experience

Ordained Minister: American Marriage Ministries

February 2024

Co-Founder & Chief Flavor Architect: Cup of Wontons
 Featured in 2024 CMU Engineering Magazine article
 Equity Researcher: Equitable and Just Greater Pittsburgh
 Co-Chair: Mechanical Engineering DEI Outreach Subcommittee
 Carolyn Commer Graduate Student Involvement Award
 Student Mentor: Carnegie Mellon Tartan Scholars Program
 Schiff Family Scholarship
 Attraction Operations Specialist: EnterTRAINment Junction
 Adult Literacy Program Volunteer: Nashville Public Library

February 2023 – December 2023
December 2022 – December 2023
January 2021 – May 2022
May 2021
August 2020 – May 2021
2018–2020
June 2017 – January 2018
January 2017 – May 2017

Students Advised

William Chen Santos MS in Biomedical Engineering, University of Miami	<i>August 2026 - Present</i>
Advaith Pradeep PhD in Electrical Engineering, University of Miami	<i>August 2026 - Present</i>
Ashutosh Kumar PhD in Electrical Engineering, University of Miami	<i>July 2026 - Present</i>
Farhad Ibrahimzade MS in Electrical Engineering, University of Miami	<i>July 2026 - Present</i>
Heman PhD in Electrical Engineering, University of Miami	<i>July 2026 - Present</i>
José Otavio Brochado Colombini PhD in Robotics, University of Toulouse	<i>January 2026 - June 2026</i>
Riccardo Fuser PhD in Robotics, University of Toulouse	<i>January 2025 - June 2026</i>
Jonas Soueidan PhD in Robotics, University of Toulouse	<i>January 2025 - June 2026</i>
Ahmed Yousef MS in Robotics Engineering, University of Genova	<i>June 2025 - April 2026</i>
Joshua Ramos BS in Electrical Enegineering, CMU	<i>February 2022 - September 2024</i>
Selvin Garcia Gonzalez MS in Mechanical Engineering, CMU	<i>February 2024 - July 2024</i>
Karla Soto Cuevas MS in Mechanical Engineering, CMU	<i>September 2023 - December 2023</i>
Sasha Kroman BS in Mechanical Engineering, CMU	<i>September 2023 - December 2023</i>
Nikhil Chinnalapatti Gopinath MS in Mechanical Engineering, CMU	<i>June 2023 - December 2023</i>
Anoushka Srivastava BS in Artificial Intelligence, CMU	<i>January 2023 - May 2023</i>